

Level-Forge

Type a sentence.

Get a puzzle level that can be won.

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Demo <https://maced78.github.io/level-forge/>

Source <https://github.com/Maced78/level-forge>

100%

of 200 test requests produced a level the game could win

0 of 200

levels were wrongly marked as winnable

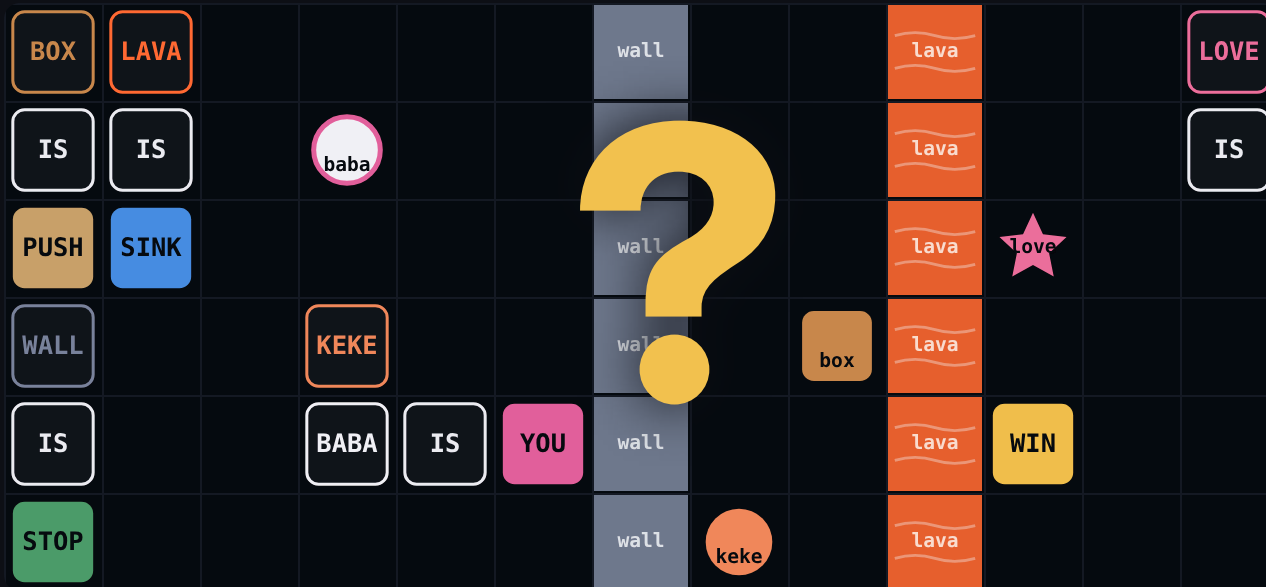
2 games

work with the same program



A hard puzzle level made from one sentence.

The problem



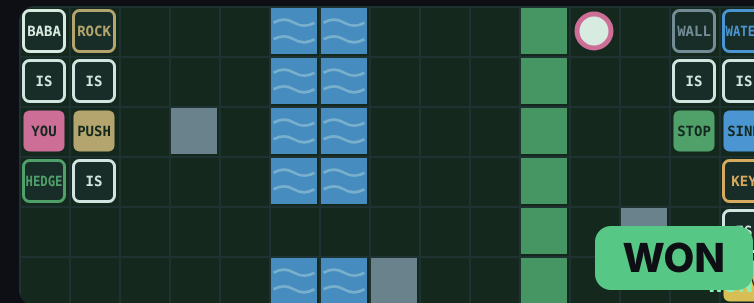
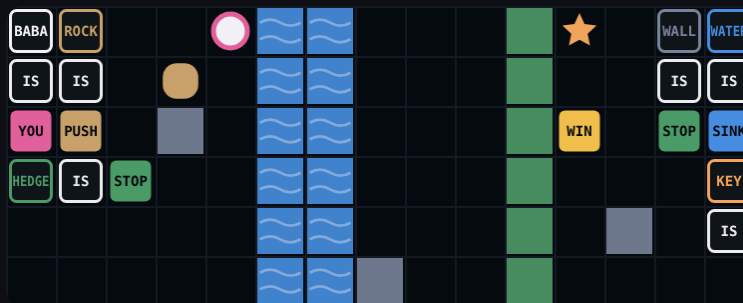
- 1 Puzzle designers draw every level by hand.
- 2 They play each one to check that it works.
- 3 Mistakes they miss reach the players.

Can this level be won? Today a person has to play it to find out.

Who has this problem: small puzzle-game studios and solo developers who ship 50 to 200 levels per game.

Our idea

"Make a hard level about pushing rocks and changing the rules."



1. You ask

In plain words.

2. It builds the level

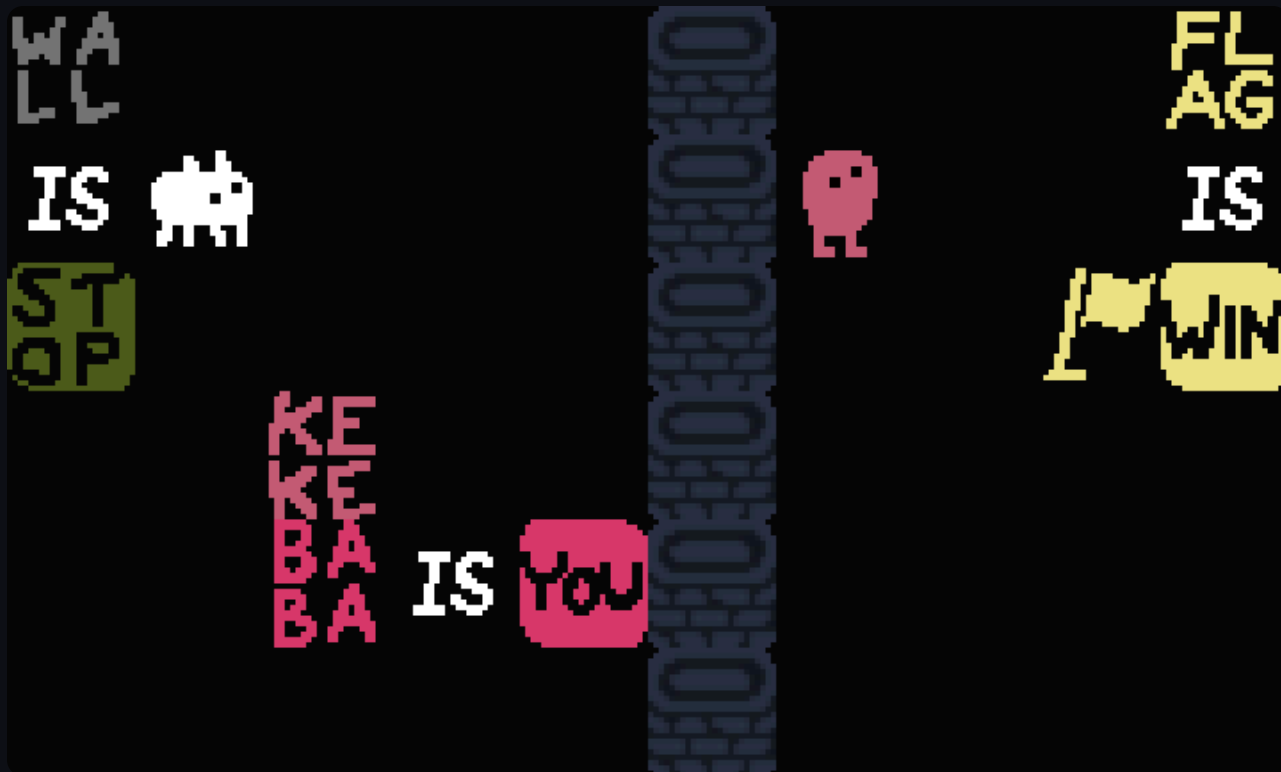
In the game's own file format.

3. The game plays it and wins

Before you ever see it.

A typical level takes about 0.8 seconds. A hard one takes about 7 seconds.

Our example game: Baba Is You



One of our generated levels, drawn with the game's own graphics.

In Baba Is You the rules are words lying on the board. Push the words and the rules change.

BABA IS YOU

you move Baba

FLAG IS WIN

touch the flag to win

WALL IS STOP

walls block you

Baba Is You was made by Arvi Teikari (Hempuli) and released in 2019. We used baba-is-auto, a free open-source copy of its rules by utilForever.

It is only our example. The same steps apply to other 2D puzzle games, and we tested a second one (Sokoban). 3D puzzle games would need extra work and are not tested yet.

An example

"Create a difficult level that tests rule manipulation, strategic object pushing and a changing win condition."



The level that came back, 15 seconds later.

- ✓ It can be won, in 47 moves.
- ✓ It cannot be won without each skill that was asked for.
- ✓ It is hard. Moving at random, a player avoids the traps 1 time in 128.
- ✓ It opens in the real game.

The computer beats its own level



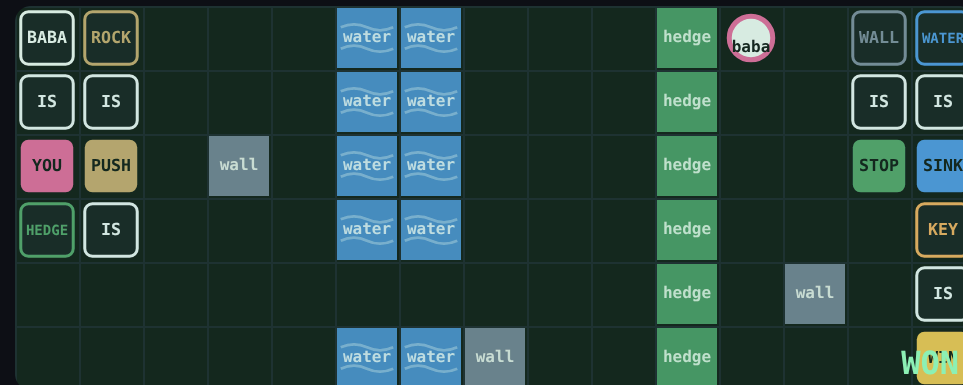
START



MOVE 16



MOVE 32



MOVE 47: WON

Every solution is played in the original game engine. You can watch it or play the level yourself at the demo link.

It refuses impossible requests

"Put the only rocks behind the water. The water can only be crossed by sinking a rock."



 Refused. You need a rock to cross the water, and the rocks are on the other side.

"Make a level with teleporters."

 Refused. This game has the word, but its code does nothing with it.

19 of 19

impossible requests refused

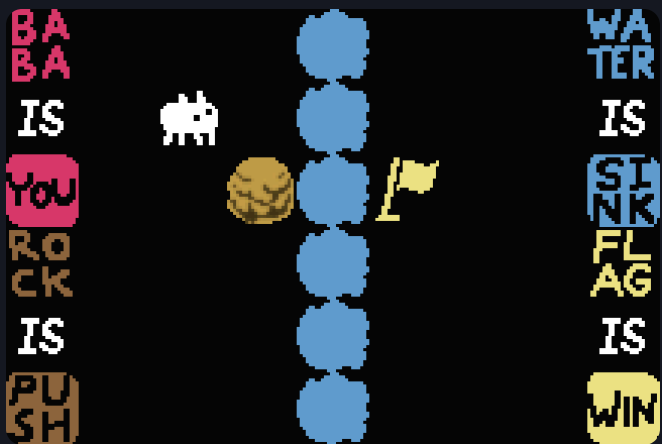
0 of 10

good requests refused by mistake

We wrote these test requests ourselves while building the checker.



It works on two different games



Baba Is You

A puzzle where the player changes the rules.

100% of 200 requests gave a winnable level



Sokoban

A puzzle about pushing boxes onto targets.

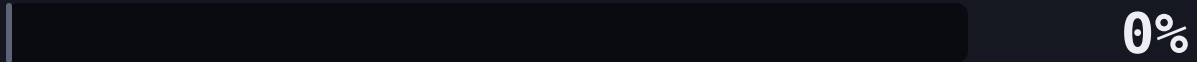
96.7% of 60 requests gave a winnable level

Both pictures are levels our program made, drawn with each game's own graphics. Adding the second game took about one hour and the main program did not change.

Test results

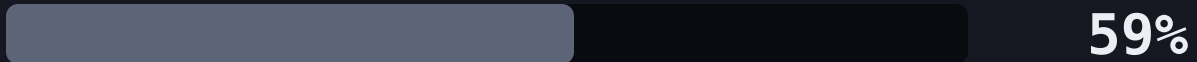
LEVELS THAT PASSED EVERY CHECK, OUT OF 200 REQUESTS

Random tiles



0%

Our builder with checking switched off



59%

Level-Forge



95.5%

Every check means: winnable, as hard as requested, and needing the requested skills.

0 of 200

levels we marked as winnable failed a second, separate replay

100,000

random moves where our fast solver and the real game agreed

60 of 60

automatic tests pass

What went wrong

- 
 Our first “hard” levels were long but easy. **Fixed:** difficulty now counts traps.
- 
 We said pushing was required. A shortcut existed. **Fixed:** our own search found it.
- 
 On the second game only 68% passed at first. **Fixed:** now 96.7%.
- 
 About 1 in 10 hard requests still misses a check. **Open:** the level is delivered with a warning.
- 
 The language model never ran. **Open.**

COMPARED WITH TODAY

	BY HAND	LEVEL-FORGE
Can it be won?	play it	proven
Shortcuts	found by luck	searched for
Difficulty	a feeling	measured

We did not time human designers or compare difficulty with real players.

Cost and next steps

0

data to collect. It only needs the game's own code.

1

ordinary computer. No graphics card, works offline.

1 hour

to add the second game.

NEXT

1

Get the language model running.

2

Add a third game with no human help.

3

Try it with a real puzzle studio.

Tested on two turn-based 2D puzzle games. Other 2D puzzle games follow the same steps. 3D puzzle games need extra work. Very large levels cannot be fully searched.

What is new



The rules come from the game's own code, and each rule is tested.



It shows the level needs the skill you asked for.



Difficulty is measured from every possible position.



It refuses impossible requests and says why.

AI chatbots can also write a level from a sentence. They work from memory and do not check the result.

Models, data and components

MODELS

- **Anthropic Claude**, as a coding assistant: read the game code and wrote the program, tests, demo and this deck with us.
- **Google Gemini**: the code is included, but it never ran. No result comes from it.
- Nothing was trained.

DATA

- Two open-source games, unmodified: **utilForever/baba-is-auto** and **xbandrade/sokoban-solver-generator** (MIT licence).
- Test requests made by our own script.
- No other data.

COMPONENTS

- Python 3, C++, pybind11
- pygame, pygame-widgets, numpy
- Playwright and Chromium
- GitHub Pages

Built after the hackathon started. Everything was made on 9 October 2026. Before the event we had only the idea and a written specification. Pages 16 to 18 hold the detailed numbers.

Thank you

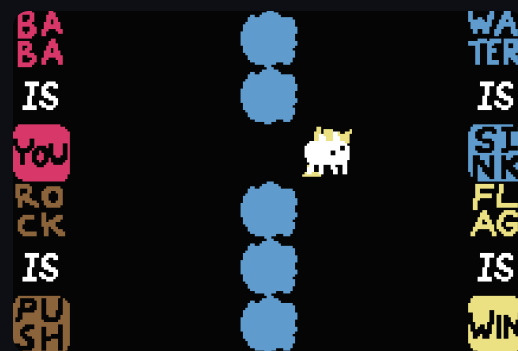
Questions are welcome.

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Demo <https://maced78.github.io/level-forge/>

Source <https://github.com/Maced78/level-forge>

Open the demo link to watch the levels being solved, or play them yourself with the arrow keys.



Two levels our program made, both already won.

What “verified” means and how difficulty is computed

FOUR LABELS

structurally valid	schema checks, no meaningless words, loaded by the original engine, a controllable object exists
proven solvable	a move sequence the original engine replays to WON (C++ and again through its Python binding)
fully validated	difficulty score inside the requested band; every requested skill exercised and proven necessary; explicit limits met
refused	no valid level can satisfy the request

Necessity by ablation. The solver is forbidden one kind of move (rule changes, object pushes, win-rule changes, control changes). If it then exhausts the state space without winning, every solution needs that skill.

DIFFICULTY WITHOUT LENGTH

score = 2 × trap bits + 4 × dead-end ratio + 2 × rule changes + 3 × sacrificed rule words

- **Trap bits:** $-\log_2$ of the chance that a player choosing at random among real moves along the solution never makes the level unwinnable.
- **Dead-end ratio:** share of reachable states that can no longer be won.
- **Bands:** easy 0–8 · medium 6–15 · hard 15+. No score if the state space exceeds the bound.

REQUESTED	MEDIAN SCORE	MEDIAN SOLUTION	FULLY VALIDATED
easy	3.5	10 moves	97%
medium	8.3	18 moves	100%
hard	20.85	44 moves	89.4%

Hard requests build dozens of candidates and deliver the hardest that passes every check. The score is our definition; it has not been compared with human ratings.

200 requests, three ways of making a level, one verifier

BENCHMARK (SEED 2026, SCRIPTS IN BENCH/, NOTHING TYPED BY HAND)			
METRIC	RANDOM TILES	BLUEPRINT, UNVERIFIED	FULL PIPELINE
Accepted by the original loader	100%	100%	100%
Solvability established (solution replayed in the engine)	0%	92%	100%
Requested skills exercised and proven necessary	0%	87.5%	99.7%
Difficulty score inside the requested band	0%	59%	95.5%
Fully validated	0%	59%	95.5%
Latency median / p90	n/a	0.001 s / 0.002 s	0.778 s / 7.406 s
"Blueprint, unverified" is the same constructor with the verifier and repair loop switched off: the gap is what verification adds. Requests come from templates over the skills the generator supports, so this measures reliability inside its vocabulary.			

- CAN THE VERIFIER BE WRONG?
- **False acceptance: 0 of 200.** Every level labelled solvable was re-played through the engine's Python binding.
 - **Model vs real engine: 117 of 117 verdicts agree** on 120 deliberately damaged levels (3 hit the engine's search bound).
 - **Differential test:** the fast search model and the original engine compared move by move on 100,000 random moves, with no disagreement.
 - **30 mechanic probes** on the real engine; validator agrees with the real loader on all 51 shipped levels.
 - **Regression suite: 60/60** checks pass.

Failures found while building it, and what each one changed

WHAT BROKE

The real engine explores only ~2,000 states per second, too slow for proofs.	Wrote a fast model of it; every solution is still replayed in the real engine and the two are differential-tested.
“Pushing is required” was false: a loose rule word can be drowned instead of a rock.	Found by the ablation search. Sink gates are now one column thicker than the number of loose words.
Early “hard” levels were long but easy: 0–1 trap points on the solution.	Replaced length with a difficulty score from the full state space; hard requests search and keep the hardest candidate.
The game's bundled viewer: tiny fixed window, scripted, no sprite for boxes or hedges.	Wrote a resizable keyboard player on the same engine.
Hard requests: 10.6% not fully validated.	Delivered with the lower label and the unmet requirement listed.
Second game, first run: only 68% fully validated.	Boxes were placed in dead corners; constructor fixed and bands calibrated; now 96.7%.
Keyword parser misses paraphrased requests.	Not fixed. The LLM reader that should replace it is written but never ran (invalid API key), so this weakness is still in every result.

COMPARED WITH HOW THE TASK IS DONE TODAY

	BY HAND	ASK AN LLM	LEVEL-FORGE
Loads in the game	yes	unchecked	checked by the real loader
Known solvable	after play-testing	no evidence	replayed in the engine
Shortcuts found	if a tester finds one	no	exhaustive ablation
Difficulty	feel	a guess	computed
Impossible request	found late	not checked	refused with reason

NOT ESTABLISHED

- Model ≡ engine is tested, not proven; “unsolvable” means with the four directional inputs.
- Two games, one version each; the difficulty score is not calibrated on human players.
- We did not time human designers.